Blastsmart app: Algorithm

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Step 1: Start

Step 2: Define variables, constant values and the put restrictions on what the user can and cannot input
Input explosive density in (g/cc)
input space to burden ratio
Input blasthole diameter in (mm)
Input bench height in (m)
Input the technical powder factor in
Define a stemming factor constraint according to environment conditions

Step 3: Define calculations using the user's inputs
Calculate the stemming length
Calculate the charge length of the explosive that is going be in each hole
Calculate the linear charge mass of the explosive
Calculate the amount of explosive that is going to be in each hole

Step 4: Expected values should be calculated and shown
Calculate the burden and show what the value is
Calculate the stemming length and show what the value is

Step 5: Layout equation to calculate the subdrill to be used in calculating the actual powder factor
Step 6: Calculate and show the actual powder factor

Step 7: Define the variables and constants for the fragmentation calculator
Input relative effective energy of explosive
Input rock factor constant
The code must recall variables instead of asking the user to re-input the same variables.

Step 8: Define calculation using the recalled inputs:
Define the Kuz-ram fragmentation model equation using the recalled user inputs

Step 9: Calculate and show the mean fragmentation size

Step 10: End